



**SAPIENZA**  
UNIVERSITÀ DI ROMA

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# Modellazione

## MIP

### Organisation of the course

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How the course is built, how every exercise is built and how the exam is built: the two-part path, the grading criteria, the typical discussion questions, the most common mistakes, and how every number quoted in the notes is reproduced.

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Online version and downloads:

[fabiofurini.github.io/modellazione-mip/](https://fabiofurini.github.io/modellazione-mip/)  
[github.com/fabiofurini/modellazione-mip](https://github.com/fabiofurini/modellazione-mip)

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# Course organization

## 1.1 The two-part path

|         | Content      | Learning objectives                                                                                                           |
|---------|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| Part I  | Modelling    | recognise a link between variables (activation, maximum, big-M, if and only if...) and prove that the model really imposes it |
| Part II | The problems | apply the links to three families of real problems and to the mixed problems, from the model to Gurobi code                   |

## 1.2 The format of every exercise (and of the exam)

Every problem of the course — and every exam question — follows the same four-question scheme:

1. **Model.** Write the MILP: variables (with their count), objective, constraints with their description, and — if two linked families of variables are involved — explain the link: what is the implication, and how the constraint (or the optimum) imposes it, in both directions.
2. **Instance.** Write the model for a small numerical instance.
3. **Heuristic.** Design a constructive algorithm that finds a feasible solution of the instance, and run it step by step to obtain an upper bound (lower bound if the problem is a maximisation).
4. **Dual.** Write the dual of the LP relaxation, for the general model and for the instance, and hand-build a feasible dual solution to obtain the bound from the other side.

The resulting  $LB \leq z(MILP) \leq UB$  is the thread running through the course: a model is not just written down; its optimum is enclosed between two numbers before it is handed to the solver.

## 1.3 The additional questions and their solutions

Every problem closes with two or three *additional modelling questions*: they change a datum or add a constraint, and ask how the model changes and what the new optimum is. They are

the part worth practising on: the model in the text is there to be read, the one for the variant has to be built.

**The solutions of the additional questions are reserved for instructors:** they live in a separate booklet, which is not published. What stays available to everyone — and it is what one needs in order to know what is expected — is the *sandwich worked out in full* on one variant of each exercise of chapters “Assignment and scheduling” and “Location and covering”: the model of the variant, a feasible constructive heuristic run step by step, a dual certificate built by hand and the bound table. It is the model answer the exam expects.

## 1.4 Grading criteria

| Dimension                                                    | Weight |
|--------------------------------------------------------------|--------|
| Correctness of the model (variables, objective, constraints) | 35%    |
| Proof of the link between the variables (both directions)    | 25%    |
| Constructive heuristic and correct execution                 | 20%    |
| Dual of the relaxation and feasible dual solution            | 20%    |

## 1.5 Typical discussion questions

- Is the link constraint aggregated or disaggregated? What difference does it make to the LP relaxation?
- Is the opposite direction of the implication imposed by the constraint or does it follow from the optimum? How is it proved?
- What is the smallest big-M that can be justified from the data?
- Does the heuristic find the optimum? How can one know without the solver?
- Is the hand-built dual optimal for the relaxation, or only feasible?

## 1.6 The most common mistakes

1. Writing an implication whose thesis is true regardless (vacuous contrapositive): before writing it, check that antecedent and consequent are both genuine.
2. Proving only one direction of an implication “imposed by the constraint” — both “if”s are always needed.
3. Confusing a relation “imposed by the constraint” with one that “follows from the optimum”: the second requires the six-step exchange argument, not just “it’s clearly worth it”.
4. Concluding “in every optimal solution” when the coefficient is only  $\geq 0$  (not  $> 0$ ): the correct conclusion is weaker, “there exists an optimum in which...”.
5. Using a huge big-M “to be safe”: it needlessly worsens the LP relaxation.
6. Forgetting that a mutual-exclusion constraint ( $A + B \leq 1$ ) does not imply that at least one of the two must equal 1.

7. Writing the dual of the relaxation with the wrong direction or sign of a dual variable relative to the direction of the primal constraint.
8. Building a feasible dual solution without checking its feasibility on all the constraints.

## 1.7 Reproducibility

```
python3 -m pip install gurobipy matplotlib pandas
python3 python/run_all.py                # regenerates data, results, figures and
    notebooks
python3 python/check_numbers.py          # checks every number quoted in the notes
```

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